



Non-Convex Gavagai

Information Bottlenecks for Formal Epistemology

Pittsburgh Formal Epistemology Workshop

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Motivation



Motivation



- $x \in X$
- $y \in Y$

Motivation



- $x \in X$

- $y \in Y$

How do we
compress X ?

Motivation



A brief plea for
naturalism



Rationality
Substantive → Procedural

Guiding Questions

X



- How can we obtain principled partitions?

Guiding Questions

X



- How can we obtain principled partitions?
- Could there be different principled partitions?

Guiding Questions

X



- How can we obtain principled partitions?
- Could there be different principled partitions?
- Do different principled partitions «just say the same thing in different ways?»

Guiding Questions

X



- How can we obtain principled partitions?
- Could there be different principled partitions?
- Do different principled partitions «just say the same thing in different ways?»
- Are principled partitions hard or soft?

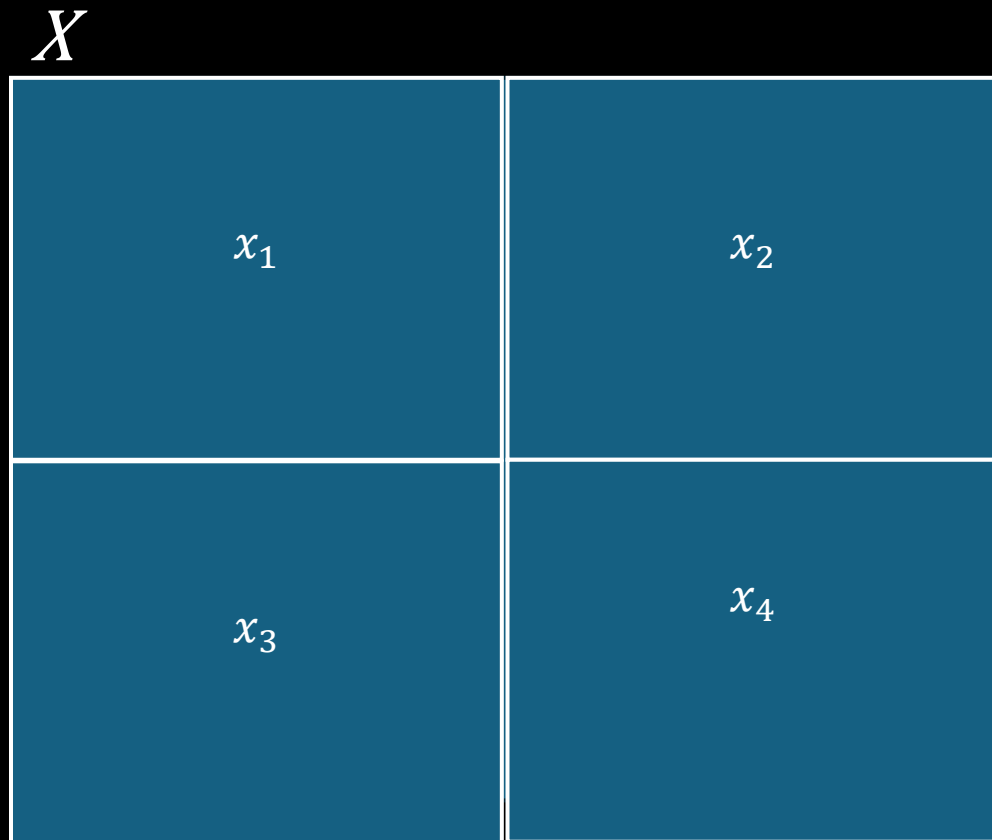
Information Theory

X

x_1	x_2
x_3	x_4

$X \sim \text{uniform}$

Information Theory*



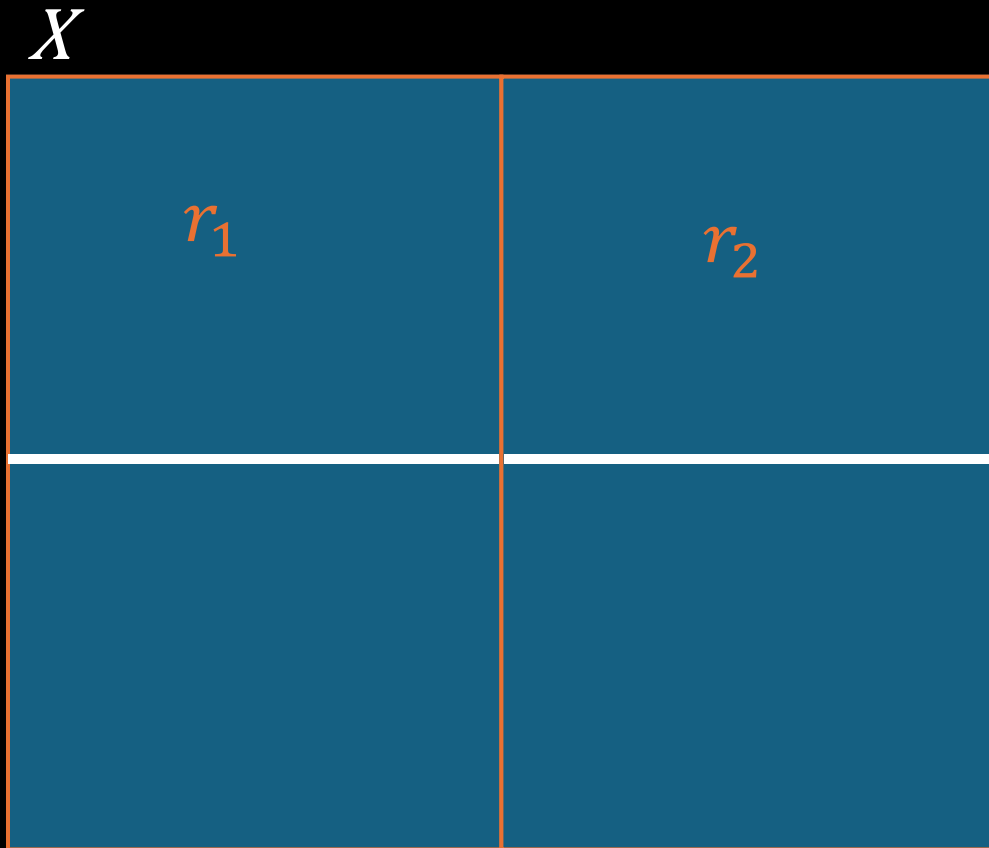
$$X \sim \text{uniform}$$

$$I(x_1) = -\log p(x_1)$$

$$H(X) = -\sum_x p(x) \log p(x)$$

*Shannon (1948): "A Mathematical Theory of Communication"

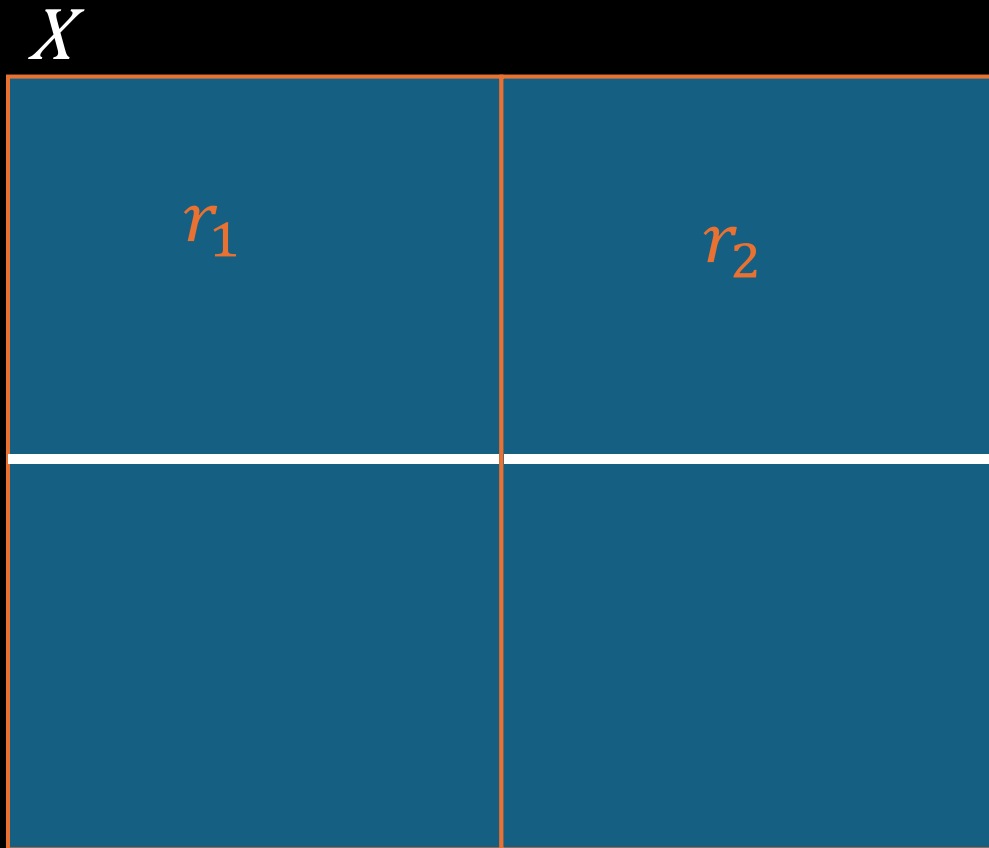
Information Theoretic Constraints



$$X \sim \text{uniform}$$
$$H(X) = - \sum_x p(x) \log p(x)$$

$$|R| < |X|$$

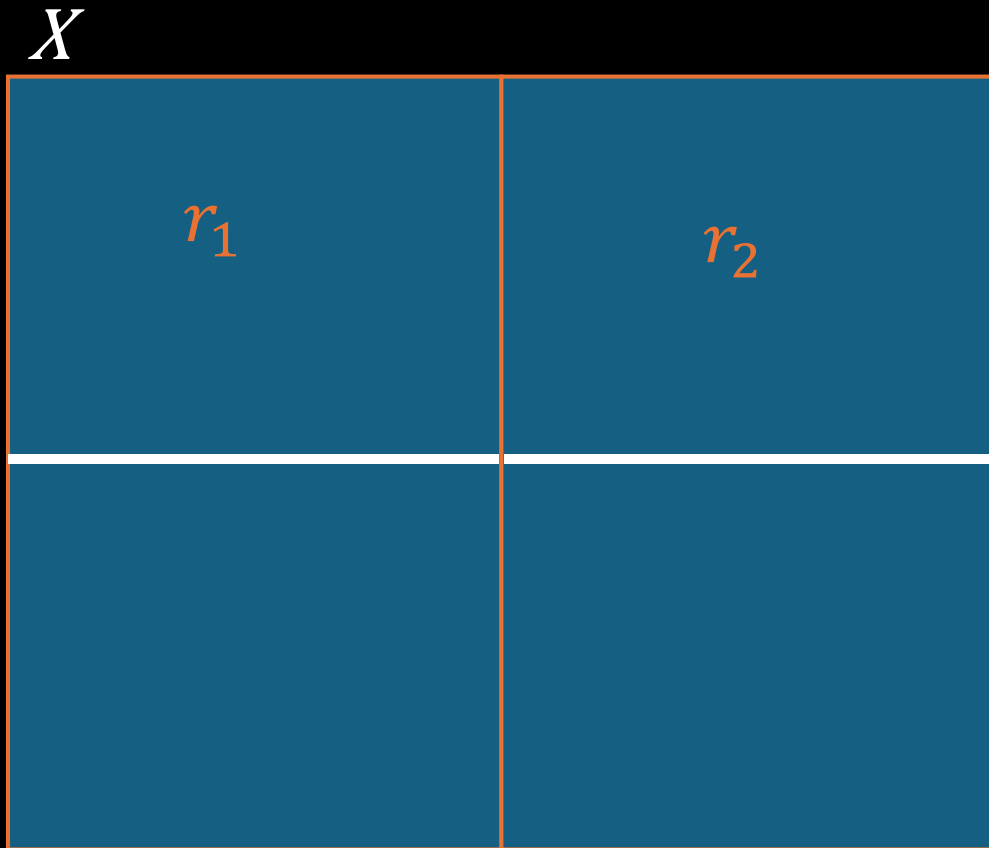
Information Theoretic Constraints



$$X \sim \text{uniform}$$
$$H(X) = - \sum_{|R|^x < |X|} p(x) \log p(x)$$

Codebook : $p(r|x)$

Information Theoretic Constraints



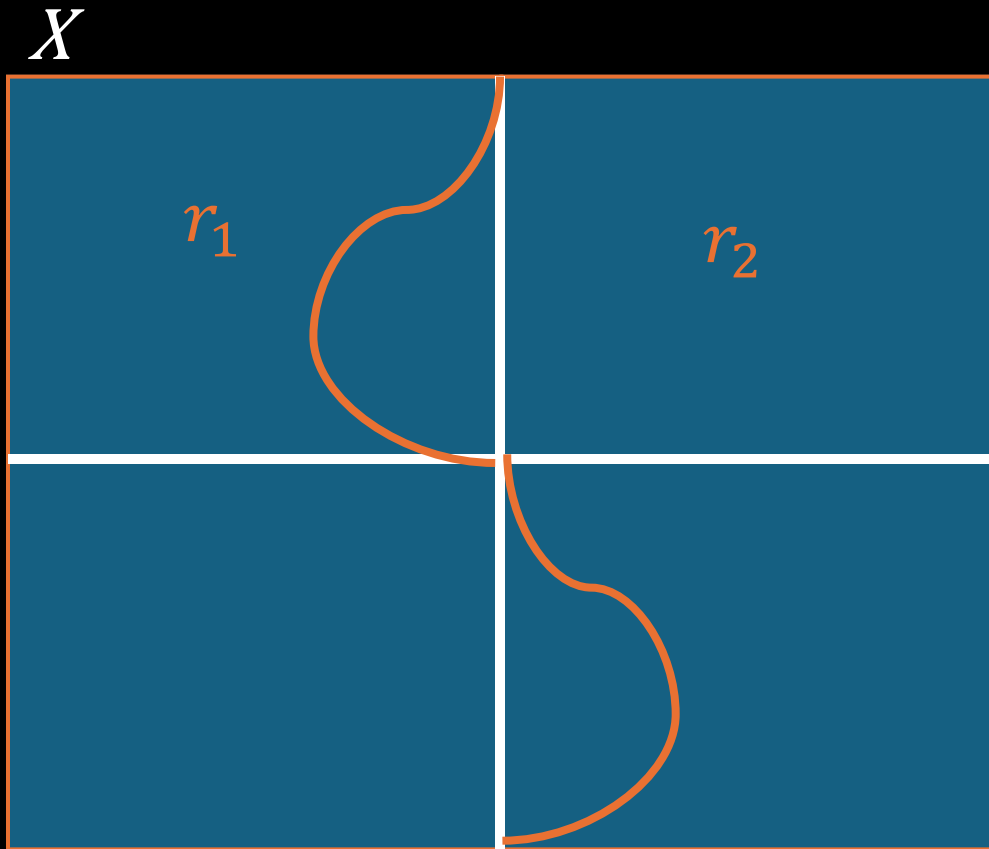
$$X \sim \text{uniform}$$
$$H(X) = - \sum_x p(x) \log p(x)$$

$$|R| < |X|$$

Codebook : $p(r|x)$

$$I(R; X) = H(X) - H(X|R) = H(R) - H(R|X)$$

Information Theoretic Constraints



$$X \sim \text{uniform}$$
$$H(X) = - \sum_x p(x) \log p(x)$$

$$|R| < |X|$$

Codebook : $p(r|x)$

$$I(R; X) = H(X) - \underbrace{H(X|R)} = H(R) - \underbrace{H(R|X)}$$

Rate Distortion*

X



*Shannon (1959): "Coding Theorems for a Discrete Source With a Fidelity Criterion"

Rate-Distortion

r_1, r_2

Cost/Neg. Utility of ‘distorting’: $d(x, r) \geq 0$

x



Rate-Distortion

$r_1 = \text{good}$, $r_2 = \text{bad}$

Cost ‘distorting’: $d(x, r) \geq 0$

X



Expected Distortion: $\mathbb{E}_{p(x,r)}[d(x,r)] = D$

Rate-Distortion

$r_1 = \text{good}$, $r_2 = \text{bad}$

Cost of ‘distorting’: $d(x, r) \geq 0$

X



Expected Distortion: $\mathbb{E}_{p(x,r)}[d(x,r)] = D$

$$R(D) = \min_{p(r|x): \mathbb{E}_{p(r,x)}[d(x,r)] \leq D} I(X; R)$$

Rate-Distortion

$r_1 = \text{good}$, $r_2 = \text{bad}$

Cost of ‘distorting’: $d(x, r) \geq 0$

X



Expected Distortion: $\mathbb{E}_{p(x,r)}[d(x,r)] = D$

$$R(D) = \min_{p(r|x): \mathbb{E}_{p(r,x)}[d(x,r)] \leq D} I(X; R)$$

Which codebook $p(r|x)$?

Loss Function

Exchange rate between D and $I(R; X)$: β

X



Loss Function

Exchange rate between D and $I(R; X)$: β

X



$$\mathcal{L}_{\mathcal{RD}}(p(r|x)) = I(R; X) + \beta \mathbb{E}_{p(x,r)}[d(x,r)]$$

$$\frac{d\mathcal{L}}{d p(r|x)} \stackrel{!}{=} 0$$

Loss Function Curvature

Exchange rate between D and $I(R; X)$: β

X



$$\mathcal{L}_{\mathcal{RD}}(p(r|x)) = I(R; X) + \beta \mathbb{E}_{p(x,r)}[d(x,r)]$$

Convex: $I(R; X) = D_{KL}[p(r, x)|p(x)p(r)]$

Loss Function Curvature

Exchange rate between D and $I(R; X)$: β

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Linear: $\beta \sum_{x,r} p(r|x)p(x)d(x, r)$

Loss Function Curvature

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Linear: $\beta \sum_{x,r} p(r|x)p(x)d(x, r)$

Minima are global

X

A solid blue square is positioned on the left side of a black background. The square is oriented horizontally and vertically. In the top-left corner of the square, the letter 'X' is written in a white, serif font.



X

Y

R

Information Bottleneck*

Y R

X

A large blue square representing the input space X .

*Tishby et al. (2000): "The Information Bottleneck problem"

Information Bottleneck

$Y \rightarrow R$

X



$p(x)$ given

$p(y|x)$ given

Information Bottleneck

$Y \quad R$

X



$p(x)$ given

$p(y|x)$ given

We want $I(R; Y)$ but no $I(R; X)$

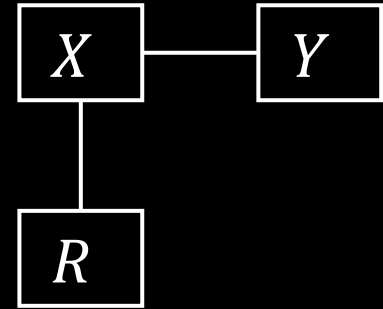
Information Bottleneck

$p(x)$ given

$p(y|x)$ given

$|X|, |Y|, |R|$

Dependency Graph



Information Bottleneck

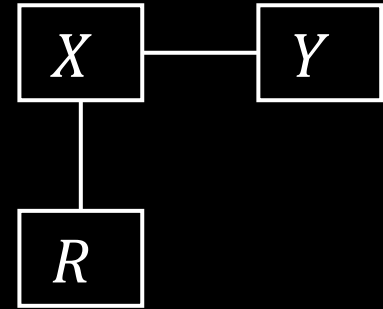
$|X|, |Y|, |R|$

$p(x)$ given

$p(y|x)$ given

$$\mathcal{L}(p(r|x)) = I(R; X) - \beta I(R; Y)$$

Dependency Graph



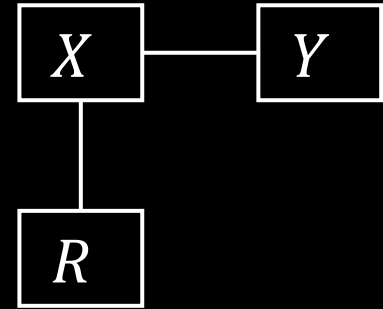
Information Bottleneck

$$|X|, |Y|, |R|$$

$p(x)$ given

$p(y|x)$ given

Dependency Graph



$$\mathcal{L}(p(r|x)) = I(R; X) - \beta I(R; Y)$$

$$\frac{d\mathcal{L}}{d p(r|x)} \stackrel{!}{=} 0$$

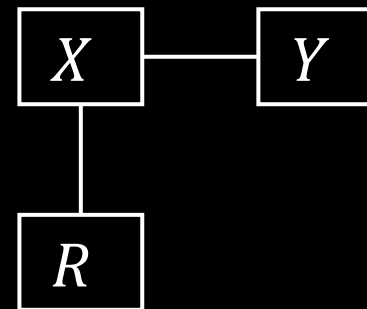
Non-convexity

$$|X|, |Y|, |R|$$

$p(x)$ given

$p(y|x)$ given

Dependency Graph



$$\mathcal{L}(p(r|x)) = I(R; X) - \beta I(R; Y)$$

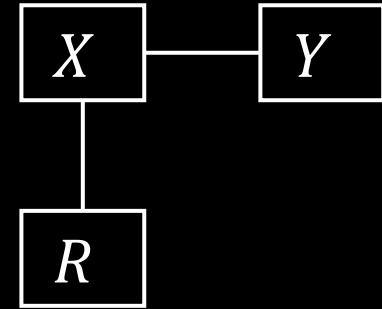
$$\frac{d\mathcal{L}}{d p(r|x)} \stackrel{!}{=} 0$$

$$p(r|x) = \frac{p(r) e^{-\beta D_{KL}[p(y|r)|p(y|x)]}}{Z(x, \beta)}$$

Non-convexity

$|X|, |Y|, |R|$
 $p(x)$ given
 $p(y|x)$ given

Dependency Graph



$$\mathcal{L}(p(r|x)) = I(R; X) - \beta I(R; Y)$$

« There is no single correct way of carving nature at its joints... nature has no joints, or rather, it has too many. » *

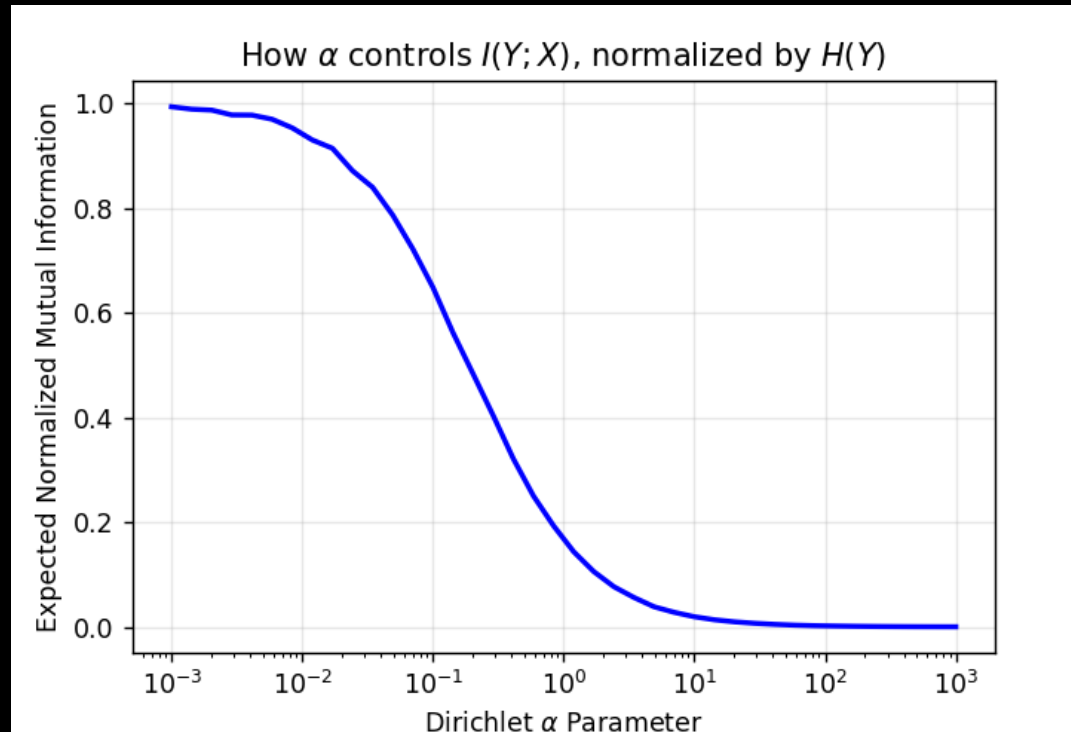
*Dupré (1993): *The Disorder of Things*

Partitions for Simulations

$$|X|, |Y|, |R|$$

Dependency X-Y

- X uniform
- $|Y| \times |X|$ Dirichlet vectors with concentration parameter α



Partitions for Simulations

$$|X|, |Y|, |R|$$

$p(x)$ given

$p(y|x)$ given

$$\mathcal{L}(p(r|x)) = I(R; X) - \beta I(R; Y)$$

Blahut-Arimoto Algorithm*

Initial guess: $p_0(r|x)$

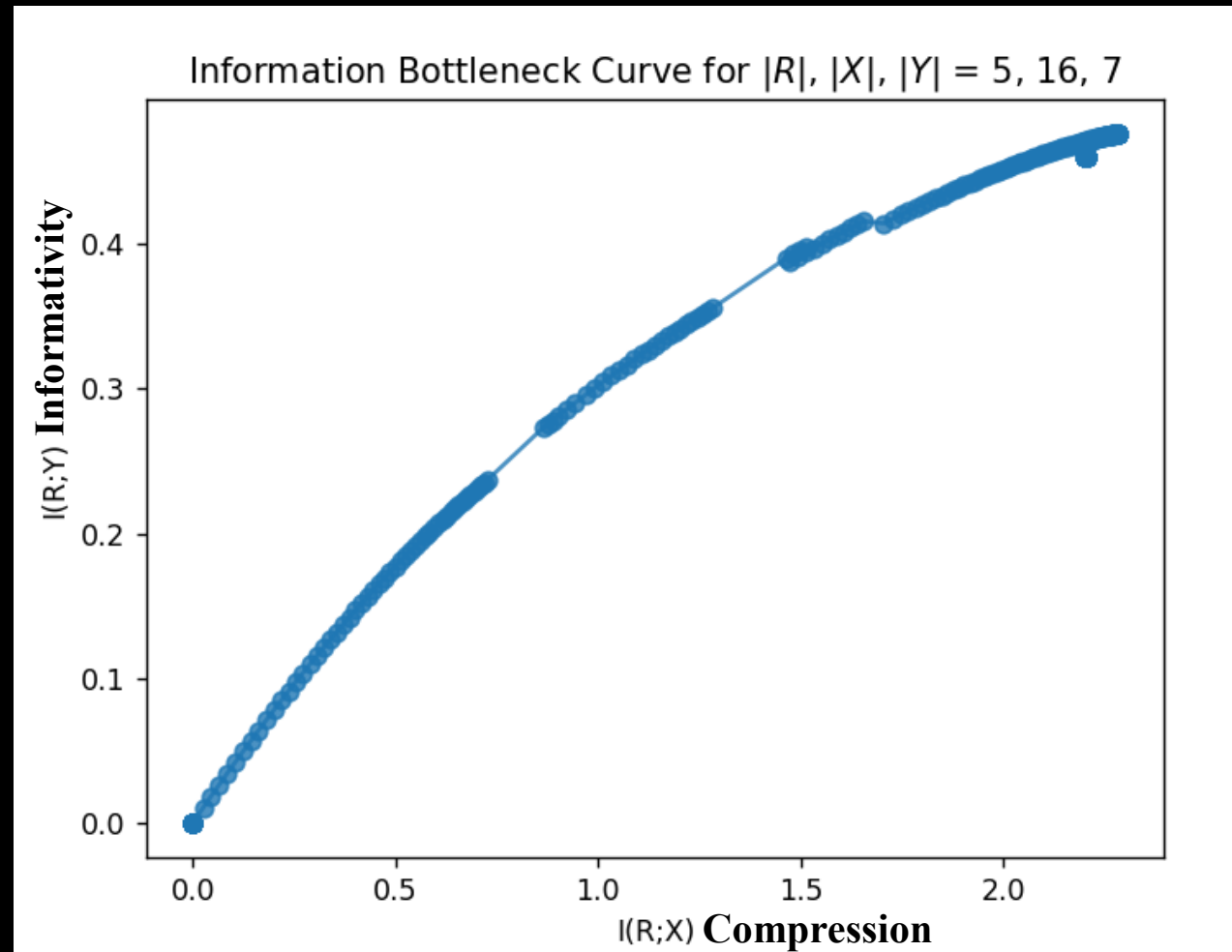
Converges to: $p(r|x), p(y|r)$

Sweep across betas

*Arimoto, S. (1972). An algorithm for computing the capacity of arbitrary discrete memoryless channels. *IEEE Transactions on Information Theory*

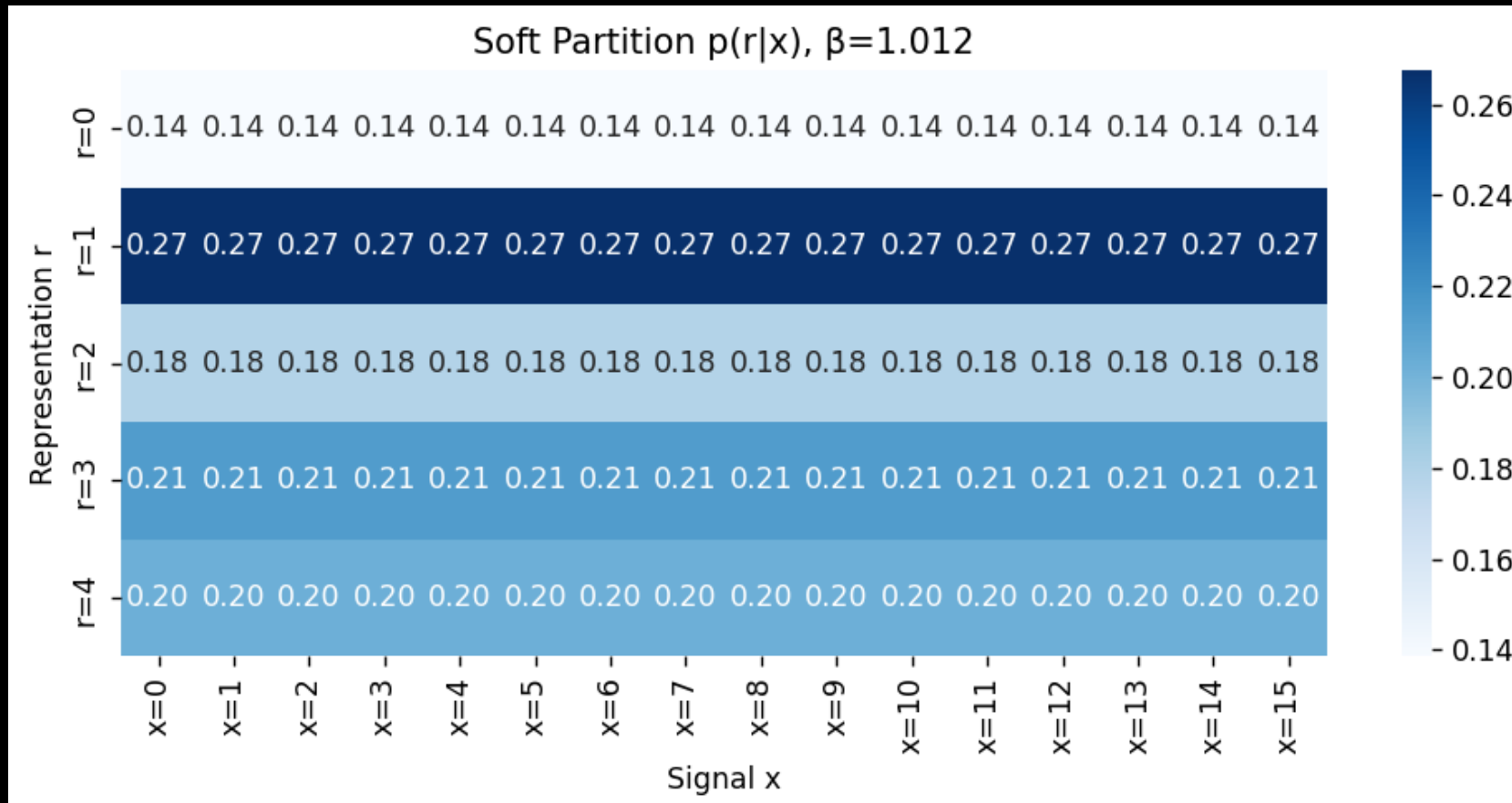
*Blahut, R. (1972). Computation of channel capacity and rate-distortion functions. *IEEE Transactions on Information Theory*

Partitions for Simulations



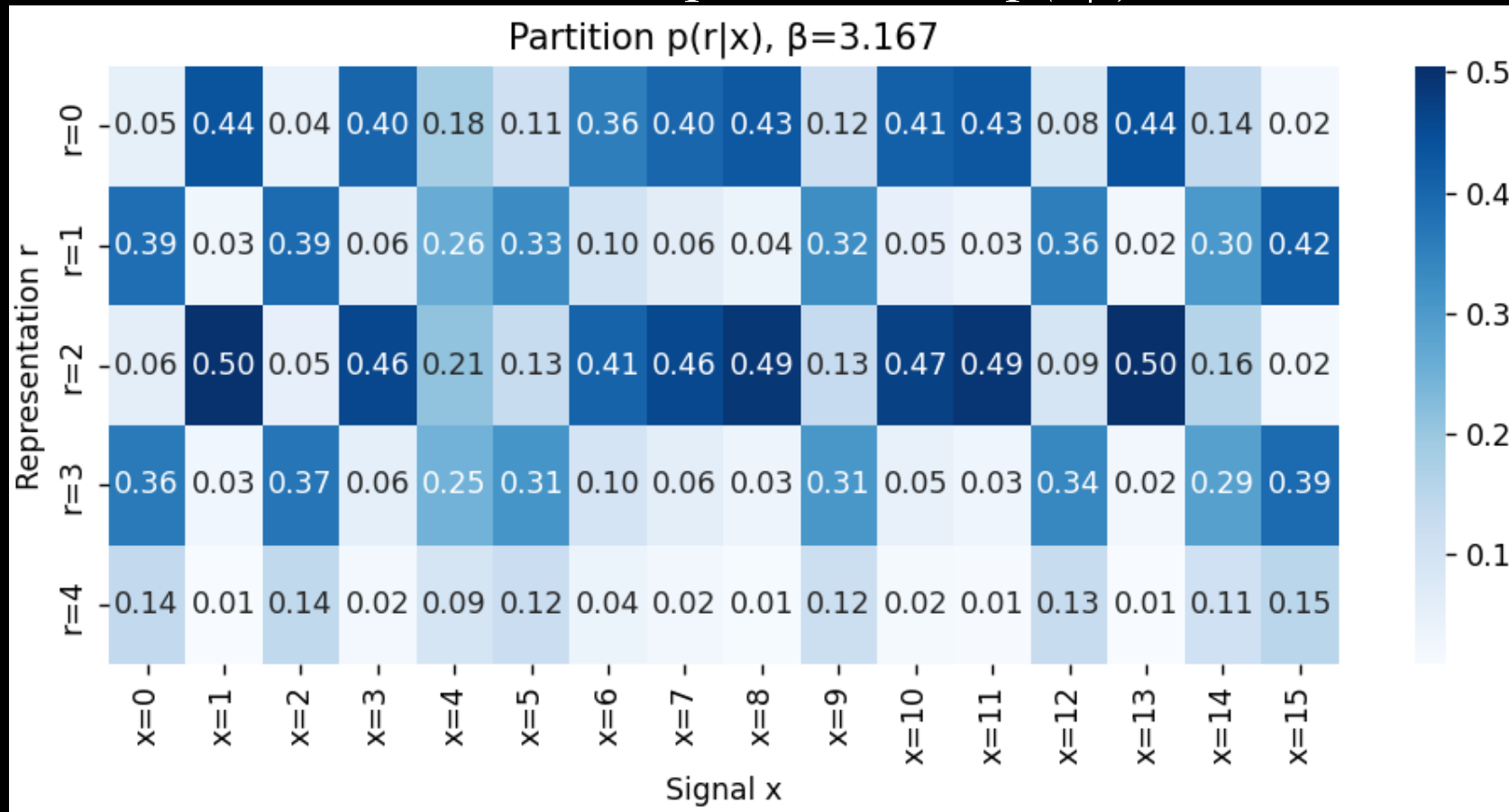
Partitions for Simulations

Partition on X per codebook $p(r|x)$



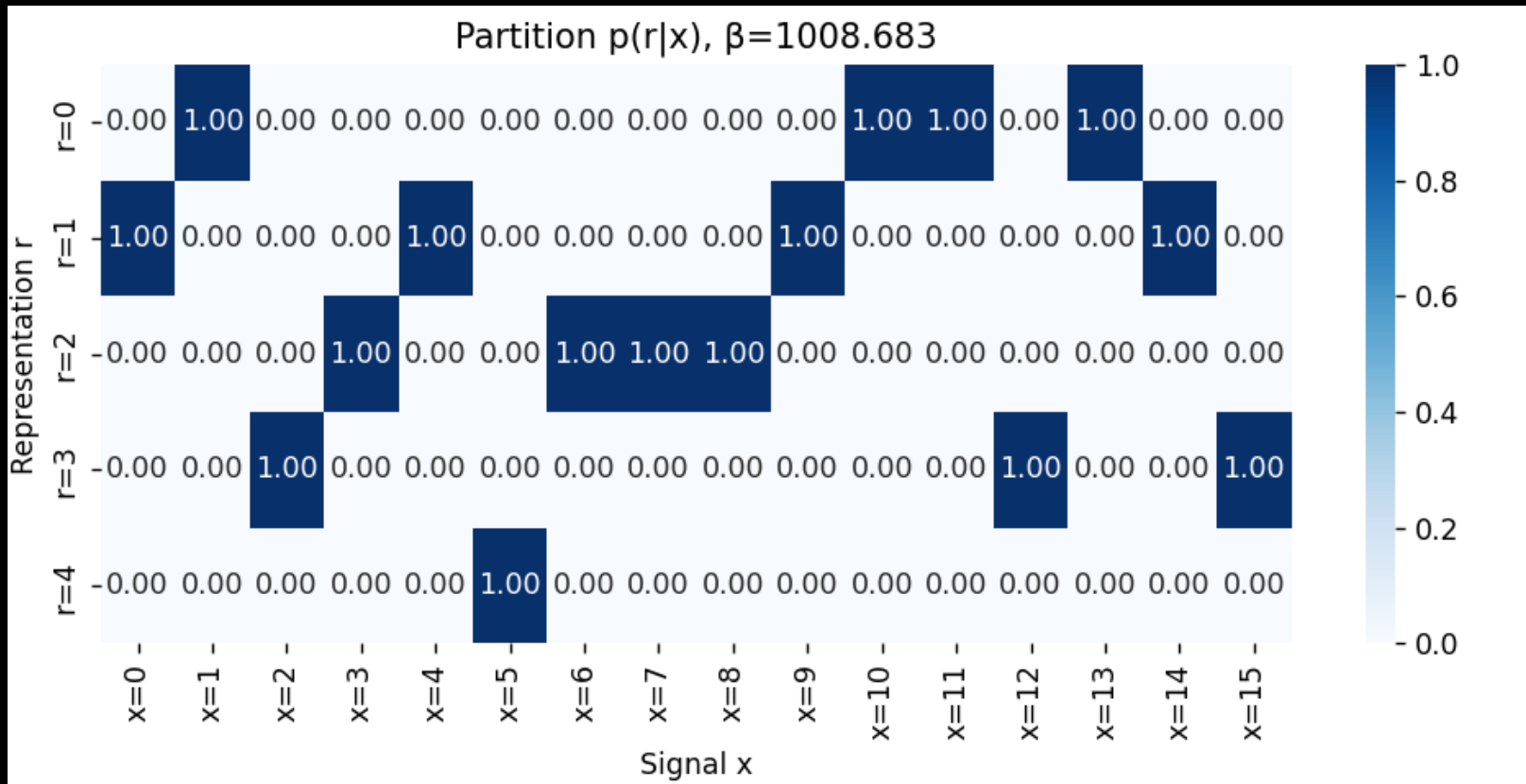
Partitions for Simulations

Partition on X per codebook $p(r|x)$



Partitions for Simulations

Partition on X per codebook $p(r|x)$



Multiple ‘Agents’

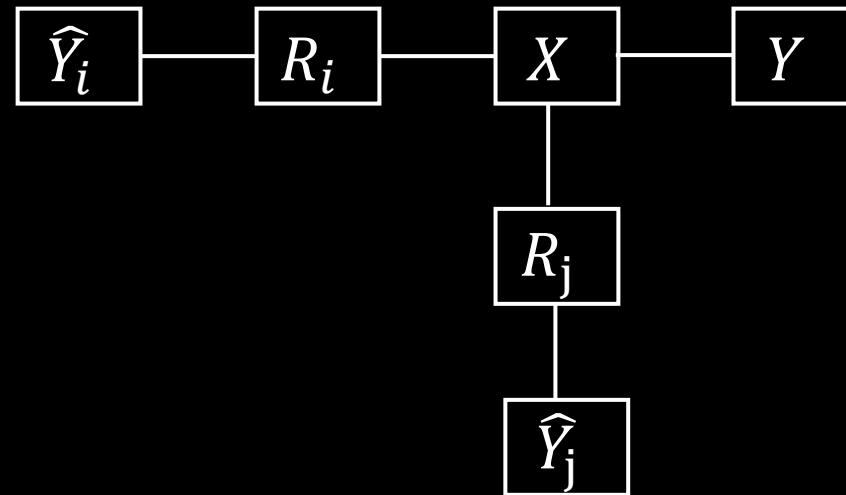
Multiple ‘Agents’

Initial guess: $p_{0,i}(r|x)$

Multiple Agents

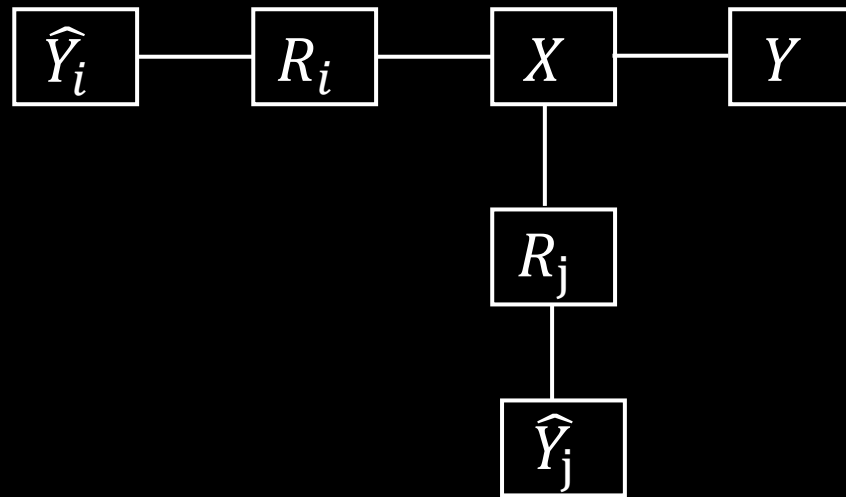
Initial guess: $p_{0,i}(r|x)$

Dependency Graph



Discord

Dependency Graph



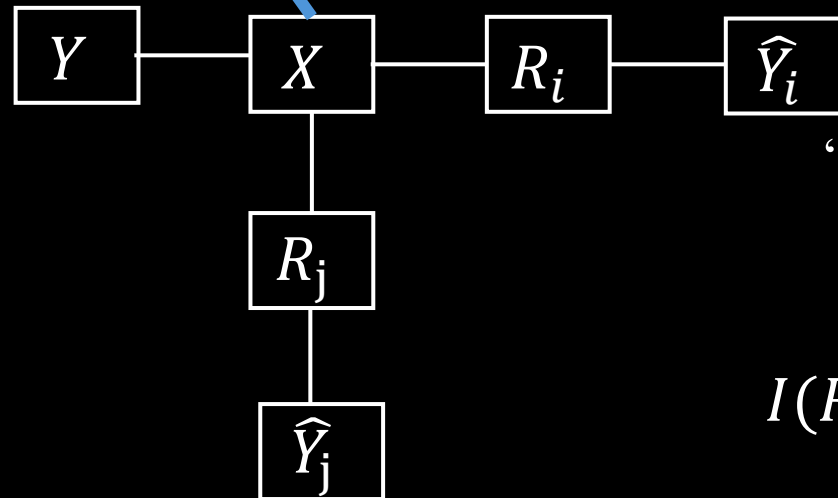
$$I(R_j, R_i) \geq I(\hat{Y}_j, \hat{Y}_i)$$

Gavagai*



*Quine, W. V. O. (1960). *Word and object*.

Gavagai

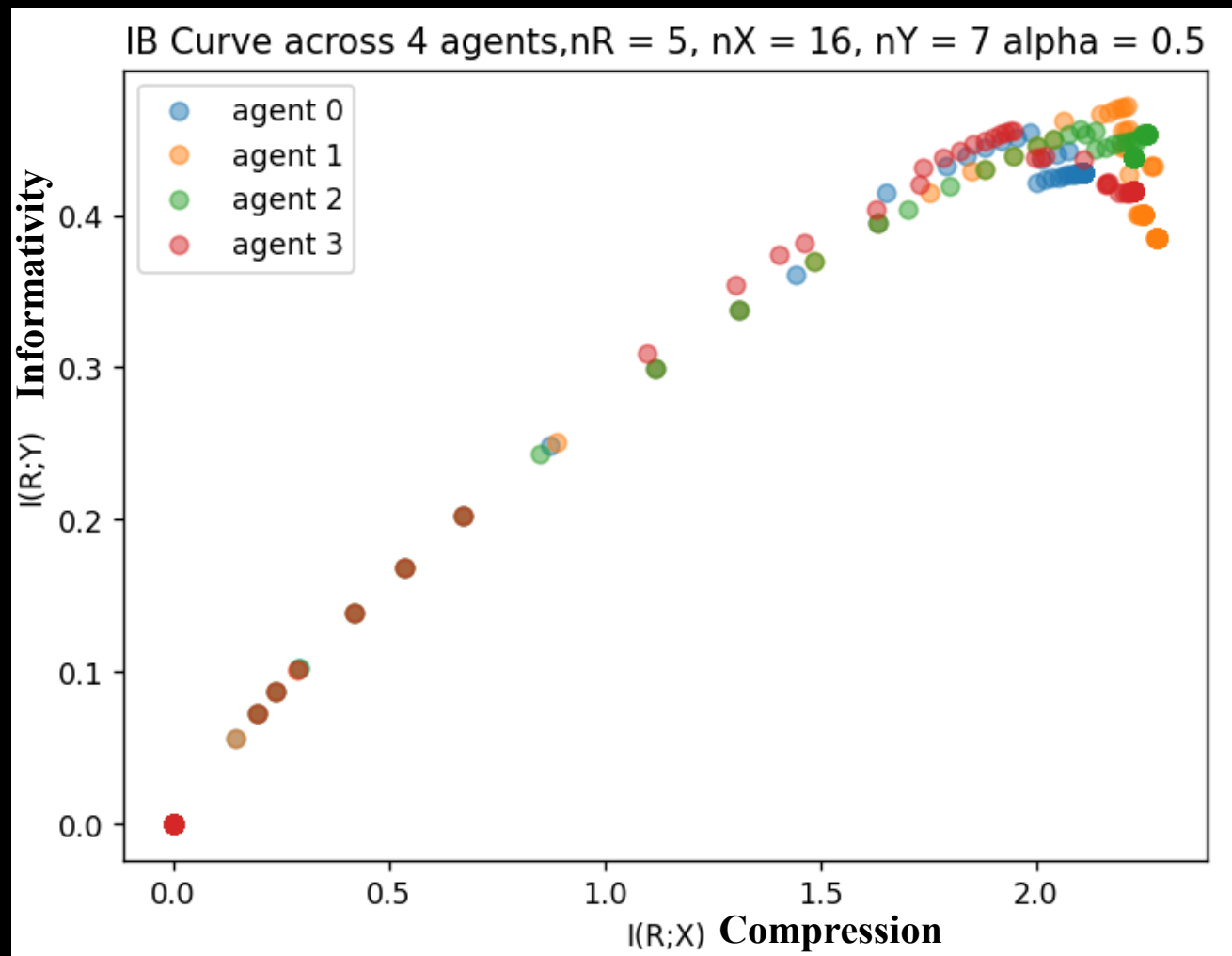
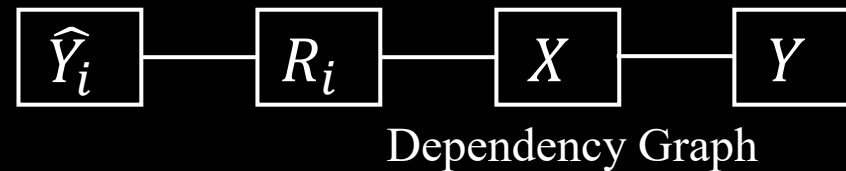


‘Gavagai’

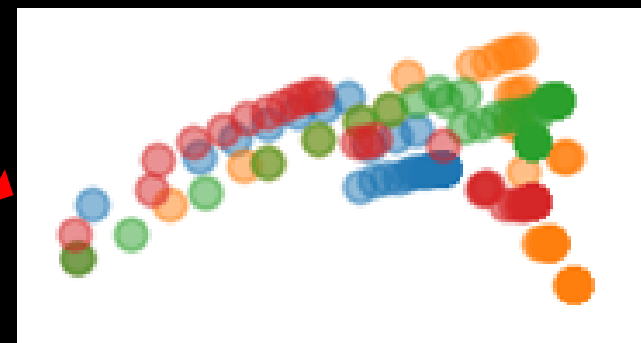
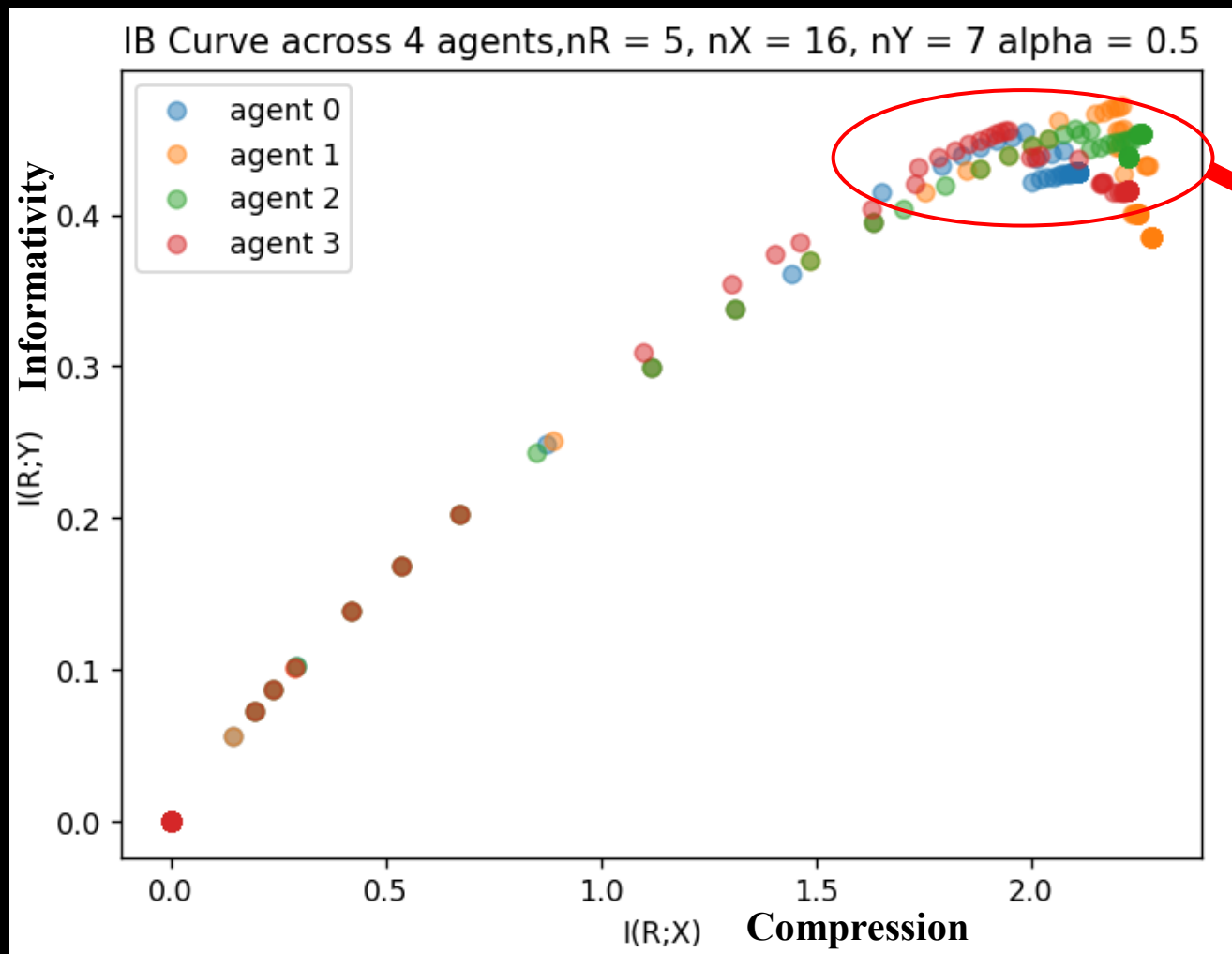
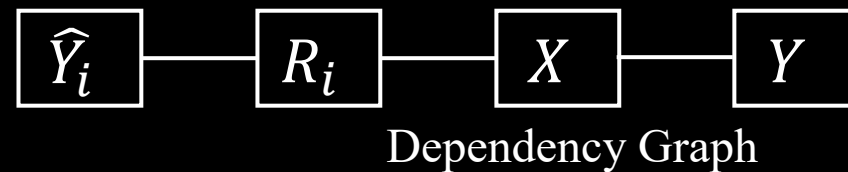
$$I(R_j, R_i) \geq I(\hat{Y}_j, \hat{Y}_i)$$

‘Lo, a rabbit’

Multiple Agents



Multiple Agents



$I(R_j; X)$: Compression

$I(\hat{Y}_j; Y)$: Informativity

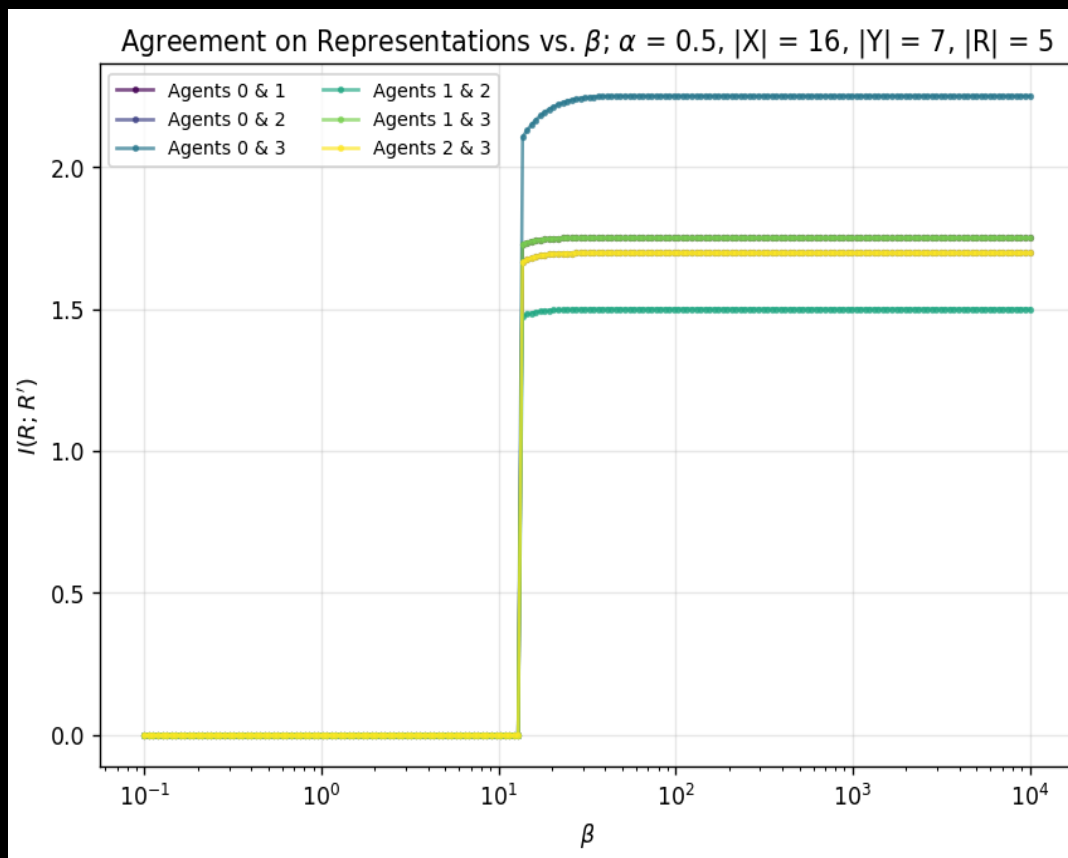
Discord

$I(R_j; R_i)$: Info about R_j given R_i

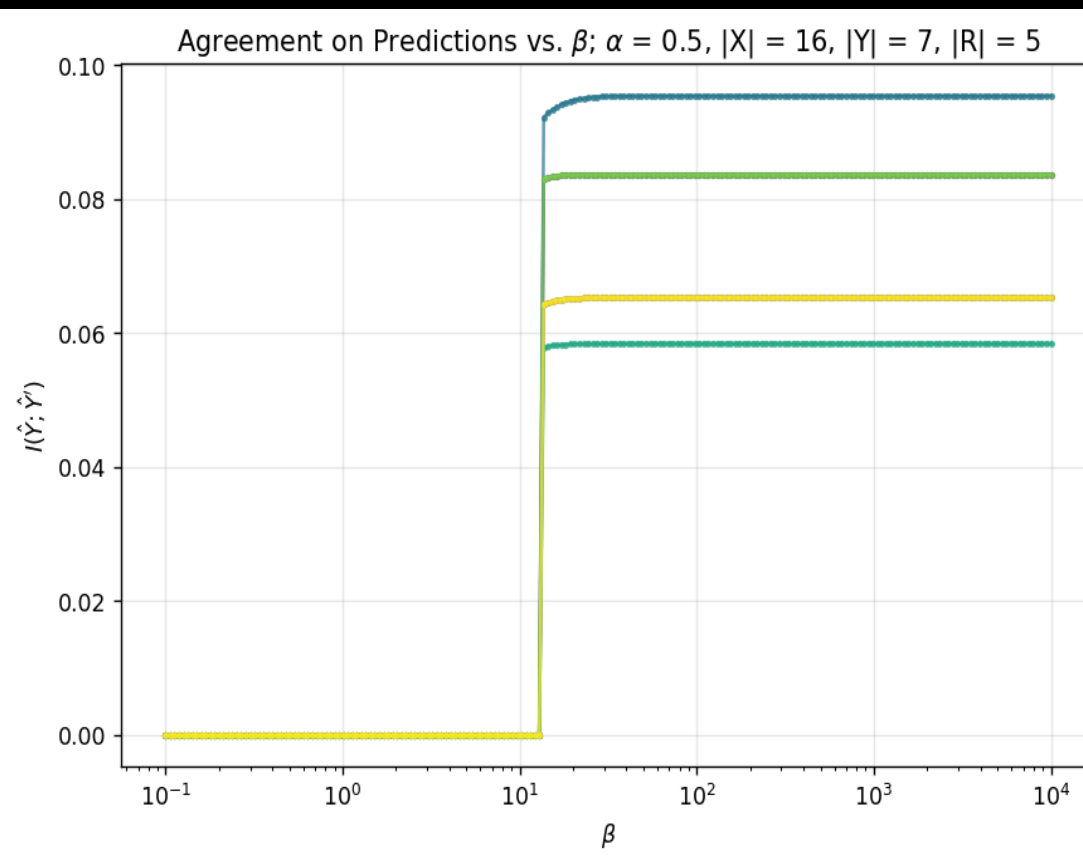
$I(\hat{Y}_j; \hat{Y}_i)$: Info about $\hat{Y}_j$ given $\hat{Y}_i$

Discord

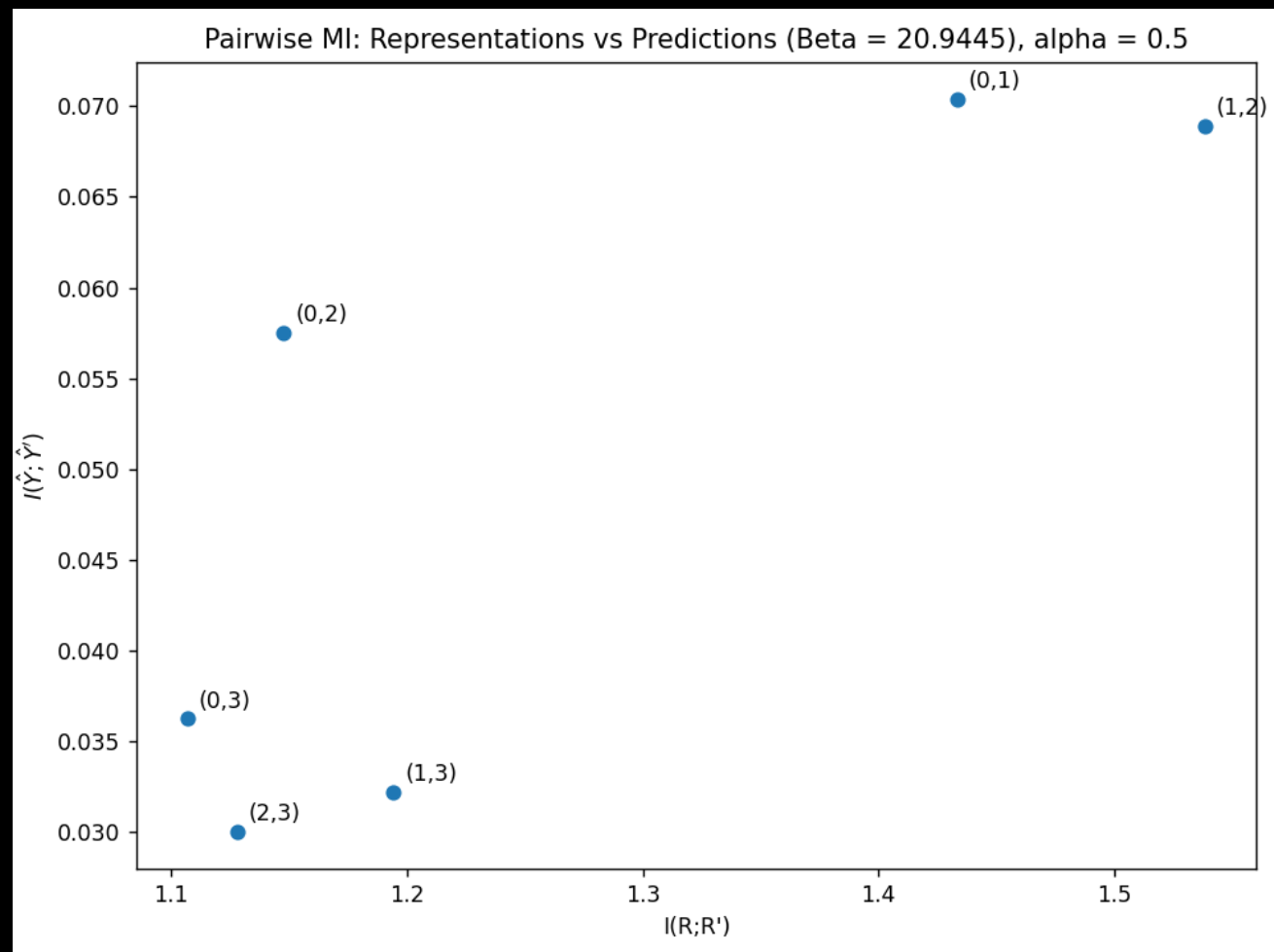
$I(R_j; R_i)$: Info about R_j given R_i



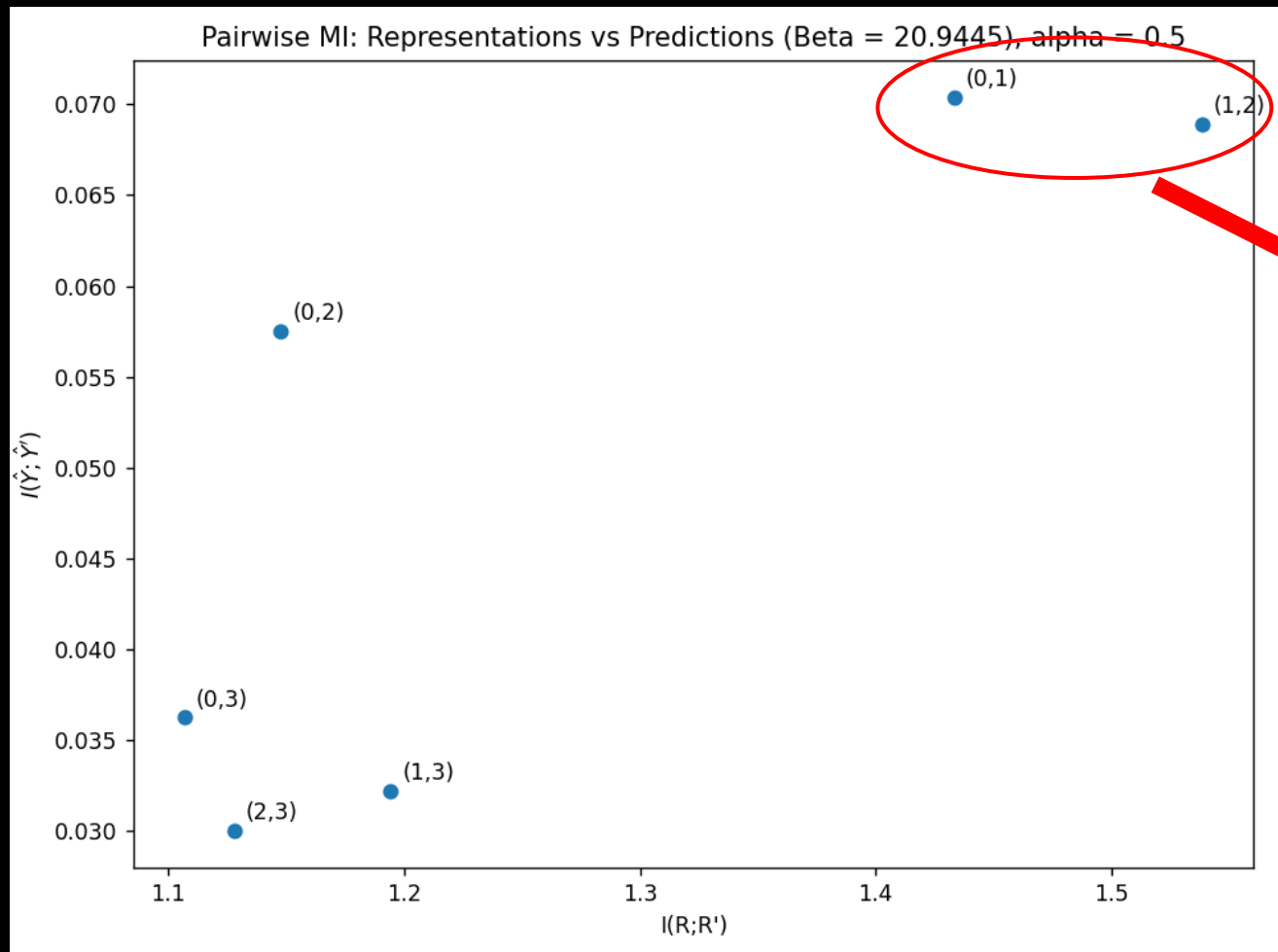
$I(\hat{Y}_j; \hat{Y}_i)$: Info about $\hat{Y}_j$ given $\hat{Y}_i$



Discord



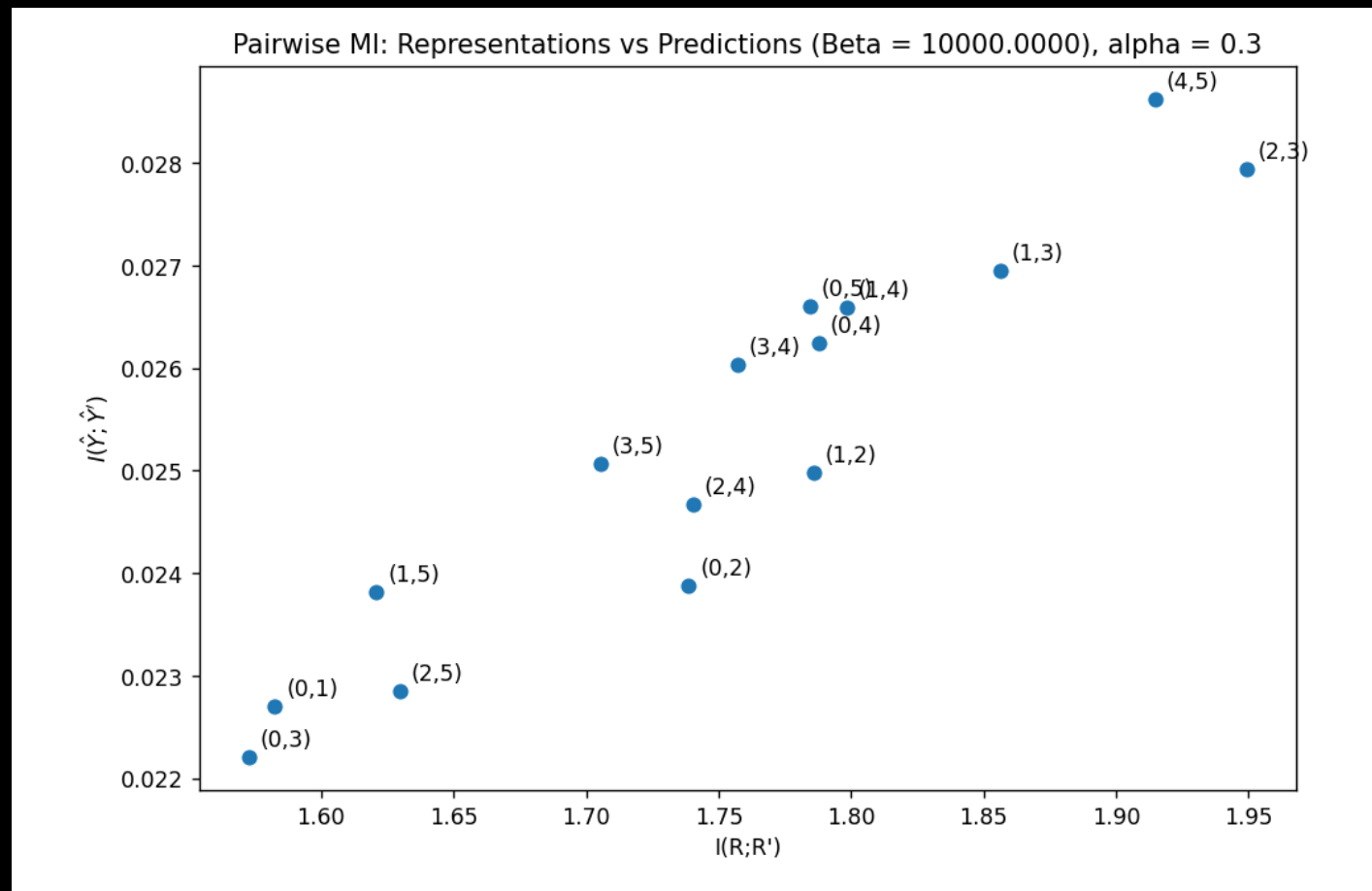
Discord



$I(R_j; R_i)$: Info about R_j given R_i

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Discord



Deterministic Representations



What genre ?

Soft v.s. Hard

«Carve nature at its joints»?*

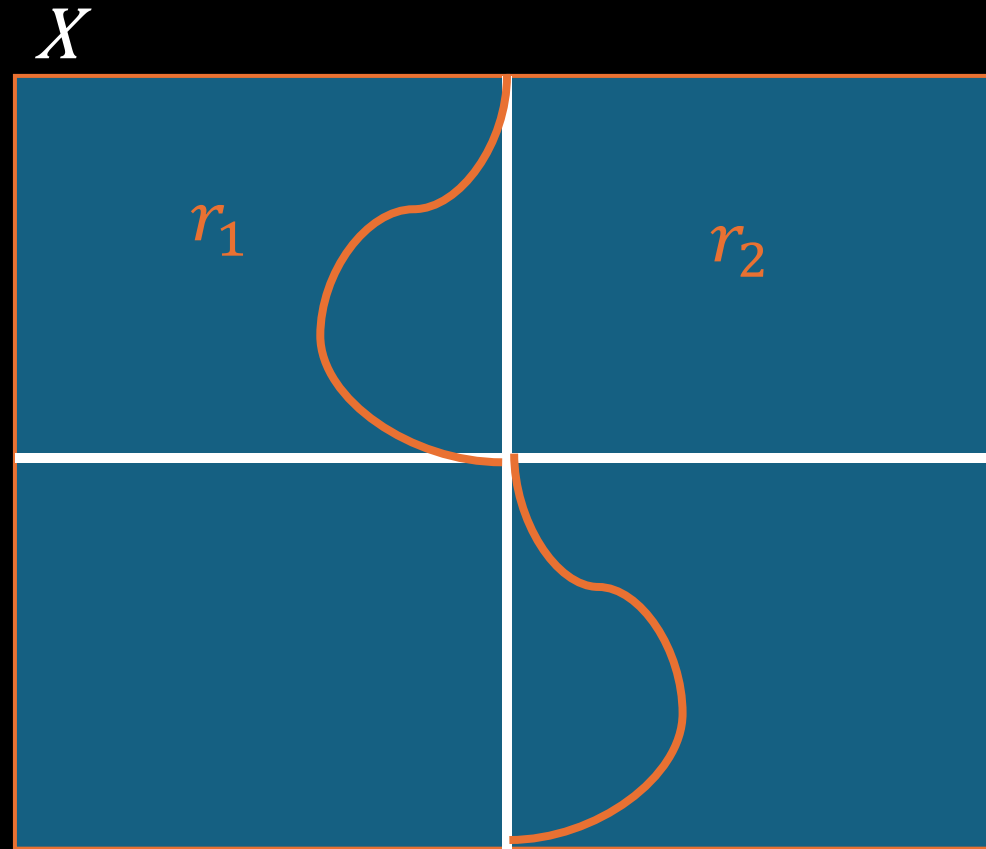
Plato's *Phaedrus**

«Most, if not all, categories do not have clear-cut boundaries»?**

Rosch (1978): *Principles of Categorization*

Soft v.s. Hard

At lower betas:

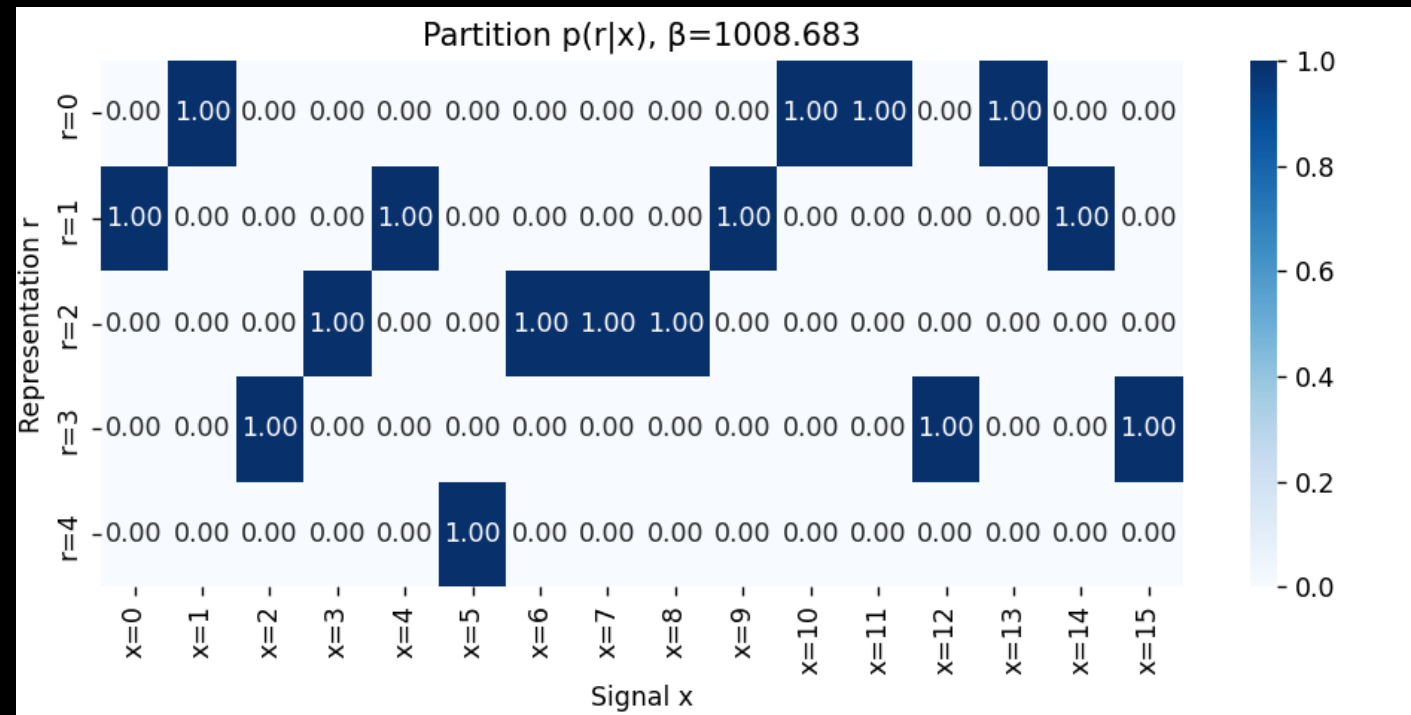


$$I(R; X) = H(X) - \underbrace{H(X|R)}_{\text{noise}} = H(R) - \underbrace{H(R|X)}_{\text{noise}}$$

Soft v.s. Hard

At $\beta \rightarrow \infty$:

Are hard partitions always optimal?



Soft v.s. Hard

At $\beta \rightarrow \infty$:

There exist optimal hard partitions.

Soft v.s. Hard

At $\beta \rightarrow \infty$:

There exist optimal hard partitions.

maximize $I(R; Y) = H(Y) - H(Y|R)$

minimize $H(Y|R)$, concave in $p(r|x)$

Global minimum obtained at extreme points of $p(r|x)$

Soft v.s. Hard

At $\beta \rightarrow \infty$:

There exist optimal hard partitions.

maximize $I(R; Y) = H(Y) - H(Y|R)$

minimize $H(Y|R)$, concave in $p(r|x)$

Global minimum obtained at extreme points of $p(r|x)$

But how common are optimal hard partitions?

Soft v.s. Hard

At $\beta \rightarrow \infty$:

$$p(r|x) = \frac{p(r)e^{-\beta D_{KL}[p(y|r)|p(y|x)]}}{Z(x, \beta)}$$

Soft v.s. Hard

At $\beta \rightarrow \infty$:

There exist optimal soft partitions when:

Soft v.s. Hard

At $\beta \rightarrow \infty$:

There exist optimal soft partitions when:

$$p(r_1|x), p(r_2|x) \in (0,1)$$

perturb: $+\epsilon$

$$\mathbb{E}_{p(y|x)}[\log p(y|r_1) - \log p(y|r_2)] = 0$$

Soft v.s. Hard

At $\beta \rightarrow \infty$:

There exist optimal soft partitions when:

Not Sure

- Symmetries in $p(y|x)$
- Uniformity in $p_0(r|x)$, deterministic $p(y|x)$

Summary

- How can we obtain principled partitions?
- Given natural constraints, IB can provide insight



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 - Given natural constraints, IB can provide insight
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- Do different principled partitions «just say the same thing in different ways?»
 - No, they genuinely capture different relevancies



Summary



- How can we obtain principled partitions?
 - Given natural constraints, IB can provide insight
- Could there be different principled partitions?
 - Yes, because of non-convexity
- Do different principled partitions «just say the same thing in different ways?»
 - No, they genuinely capture different relevancies
- Are principled partitions hard or soft?
 - Under compression constraints, soft is smart; If accuracy is imperative, hard is common

Outlook

- Create a large simulated data set
- Justify procedural rationality: annealing, many trials, many betas?
- Details on how likely soft partitions are
- Combining codebooks, combined updating

